

A Binary-Catalyst Call: When a Falling Hazard Ratio Cancels Theta

SELLAS / SLS — GPS REGAL readout

2026-06-02

Setup

A binary catalyst (the REGAL topline) lands at a random month τ with arrival density $f(\tau)$ and survival $S(t) = P(\tau > t) = \int_t^\infty f$. Define the *readout hazard*

$$h(t) = \frac{f(t)}{S(t)}.$$

At the readout the stock jumps to U (success) with probability $p(\tau)$ or to D (failure) otherwise. We hold a call struck at K , with $D < K < U$, paying $\text{PAY} = U - K$ on success and 0 on failure; r is the risk-free rate. Pre-readout diffusion is ignored (appropriate for a binary event).

Conditional on *no readout yet* at month t , the call is the discounted, probability-weighted payoff over every time $\tau \in (t, T]$ it could still fire before expiry T :

$$C(t) = \frac{1}{S(t)} \int_t^T f(\tau) p(\tau) \text{PAY} e^{-r(\tau-t)} d\tau$$

where $p(\tau)$ rises with τ because a slower-accruing trial implies a lower hazard ratio (see last section); we model $p(\tau) = \text{clip}(p_0 + \alpha\tau, 0, 1)$, so $dp/dt = \alpha > 0$.

The exact dynamics: one ODE

Differentiate the call. Writing $C(t) = \text{PAY} e^{rt} G(t)/S(t)$ with $G(t) = \int_t^T f p e^{-r\tau} d\tau$, so $G'(t) = -f(t)p(t)e^{-rt}$, and using $S'(t) = -f(t)$,

$$\frac{d}{dt} \left(\frac{e^{rt}}{S(t)} \right) = \frac{e^{rt}}{S(t)} (r + h(t)),$$

the two terms collapse to a single linear ODE:

$$\frac{dC}{dt} = (r + h)C - hp\text{PAY} = \underbrace{rC}_{\text{carry}} + \underbrace{h[C - p\text{PAY}]}_{\text{jump term}}$$

Numerical verification (exact, not approximate). At $t = 2$ ($h = 0.406/\text{mo}$, $C = 5.28$, $p = 0.50$, $\text{PAY} = 10$):

$$(r + h)C - hp\text{PAY} = 0.4093(5.28) - 0.406(5.0) = 2.161 - 2.030 = +0.131/\text{mo},$$

matching the numerically differentiated $dC/dt = +0.131$. At $t = 4$ the ODE gives -0.004 versus an actual -0.002 . The ODE is the dynamics.

The two forces

The jump term is $h[C - p\text{PAY}]$. Split the gap $C - p\text{PAY}$ into a miss-drag and a rising- p lift, with $q(t) \equiv S(T)/S(t)$ the probability the readout *misses* expiry given none yet:

$$\frac{dC}{dt} \approx \underbrace{-hp\text{PAY}q}_{\text{THETA (miss-drag)}} + \underbrace{\alpha\text{PAY}(1-q)}_{\text{RISING-}p \text{ (falling HR)}} + rC.$$

t (mo)	h/mo	$q = S(T)/S(t)$	THETA	RISING- p	net dC/dt
0.5	0.03	0.010	-0.00	+0.25	+ climb
2.0	0.41	0.014	-0.03	+0.25	+ climb
4.0	0.74	0.047	-0.19	+0.24	$\approx$ o peak
5.0	0.83	0.102	-0.49	+0.22	- decay
7.0	0.95	0.617	-3.68	+0.10	- collapse

The cancellation condition

Setting $dC/dt = 0$ (dropping the small rC) gives the flat-call locus two equivalent ways:

$$C(t) = p(t)\text{PAY} \cdot \frac{h}{r+h} \approx p(t)\text{PAY} \quad \Longleftrightarrow \quad q^*(t) = \frac{\alpha}{\alpha + ph(t)}$$

The call is flat exactly when its pre-readout value equals its *fire-now* value $p\text{PAY}$. It **climbs while** $q(t) < q^*$ and **decays once** $q(t) > q^*$. With $q^* \approx 0.06$ throughout, the crossover is at $t^* \approx 4$ months — the observed price peak ($C \approx \$5.47$).

What “HR cancels theta” really means

The loose phrasing is imprecise; the exact picture from the split is:

- **Early, theta is already ~ 0** — not because rising- p cancels it, but because $q \approx 0.01$: a readout this likely to beat a Jan-2027 expiry has almost no decay to cancel ($-h p q \text{PAY} \approx -\$0.03/\text{mo}$).
- **The falling HR adds a positive drift** ($\approx \alpha \text{PAY} \approx +\$0.25/\text{mo}$) on top of ~ 0 theta, so the call *climbs* rather than merely holding flat.
- **True one-for-one cancellation occurs only in a narrow band at the peak** ($t^* \approx 4$ mo), where theta has grown to $-\$0.19/\text{mo}$ and finally matches the $+\$0.24/\text{mo}$ lift.

α is pinned by the science, not a free knob

The lift slope is

$$\alpha = \frac{dp}{dt} = \frac{dp}{d\text{HR}} \cdot \frac{d\text{HR}}{dt}.$$

The ABC fit to the public death trajectory gives a drift $d\text{HR}/dt \approx -0.07$ over the accrual window (median interim **HR 0.71** $\rightarrow$ final **0.64** as deaths accrue slowly); the two-state PoS map turns this into $dp/dt \approx +0.025/\text{mo}$. That *negative* $d\text{HR}/dt$ is the entire source of the RISING- p term.

Counterfactual. Set $\alpha = 0$ (HR does *not* fall as the trial drags). Then the lift vanishes and the split reduces to $dC/dt = rC - h p q \text{PAY} \leq 0$ for all t : **no climb, no hump — the call only ever decays**. The hump exists *iff* HR falls with time.

The rigorous $p(\tau)$: a filtered ABC posterior

The linear ramp above is a teaching device. The defensible object is the **filtered posterior**

$$p(\tau) = P(\text{final logrank HR} < 0.636 \mid \mathcal{D}_{\text{now}}, \text{80th death at month } 64 + \tau),$$

obtained by accepting ABC draws (flat over the GPS cure-tail; wide priors on BAT median OS, dropout, differential transplant, Weibull shape) that reproduce the *public death trajectory through today* — 60th death at month 46, 72 deaths by month 58, 78 by month 64, interim passed futility — then binning the survivors by the simulated arrival month of the 80th death and reading off the success rate. BAT is marginalized out; sampling noise and the cure-tail non-proportionality are intrinsic.

τ (mo)	0	1	2	3	4	6	8	12
$p(\tau)$ filtered	0.40	0.43	0.46	0.49	0.53	0.57	0.62	0.71
naive $0.45 + 0.025\tau$	0.45	0.48	0.50	0.52	0.55	0.60	0.65	0.74

Three corrections to the ramp:

- **Slope — vindicated.** Local fit over $\tau \leq 7.5$: $p(\tau) \approx 0.405 + 0.0275 \tau$. The naive $+0.025/\text{mo}$ was essentially right.
- **Level — too high.** Real intercept ≈ 0.40 vs naive **0.45**; the readout-timing-weighted **PoS today is 0.48, not 0.52** — and it is now *derived from the trajectory*, not assumed.
- **Shape — convex and saturating.** $p(\tau)$ accelerates mildly, then bends toward a ~ 0.71 ceiling; the line has neither feature (hence its **clip**).

Feeding the filtered $p(\tau)$ through the *same* option integral:

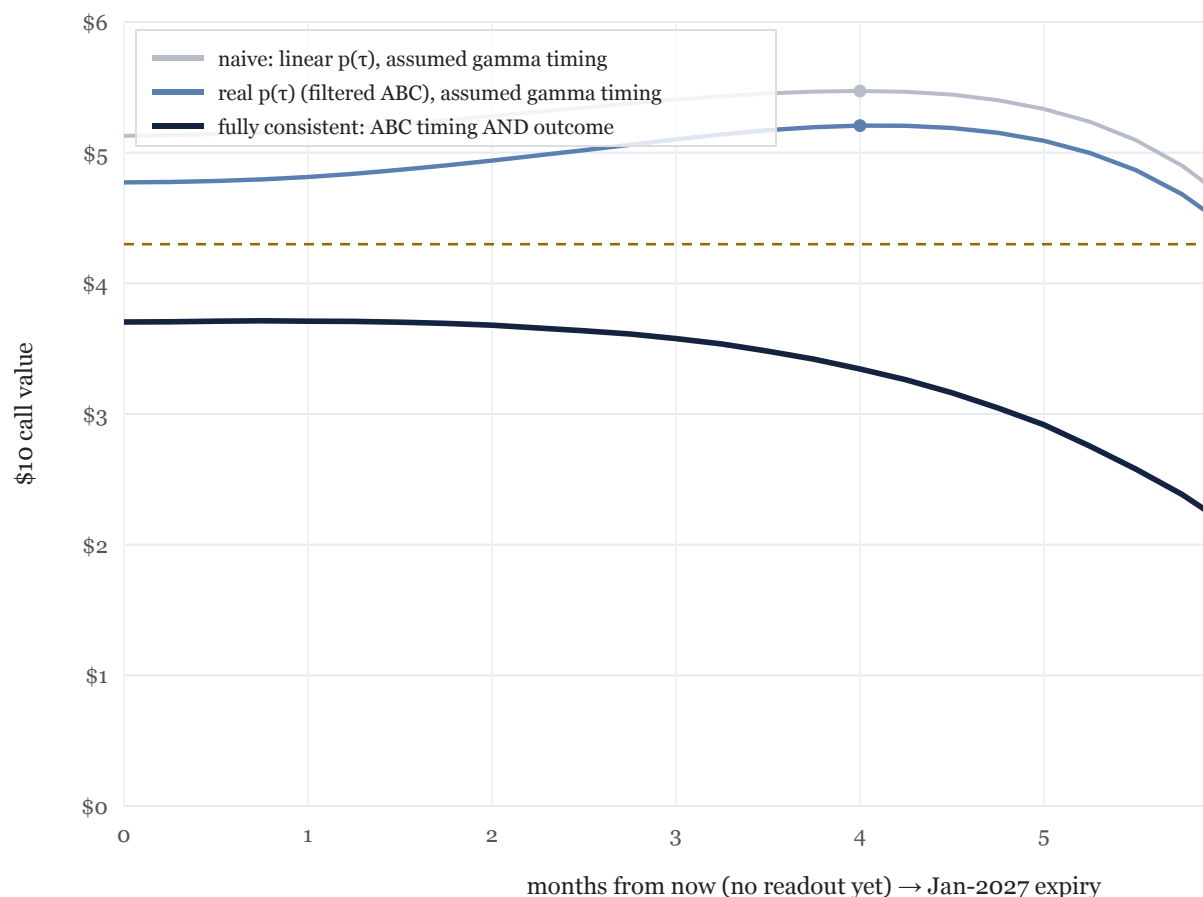
t (mo)	0	2	4 (peak)	6	7
C_{naive}	5.13	5.28	5.47	4.64	2.41
C_{real}	4.77	4.94	5.21	4.43	2.31
diff	-0.36	-0.34	-0.26	-0.21	-0.11

What survives and what moves. The physics — the hump, $dC/dt = 0$ at $q^* = \alpha/(\alpha + ph)$, the 4-month no-bleed horizon — is **robust**: it rides on the *slope* $\alpha > 0$, which the ABC confirms. The peak stays at $t^* \approx 4$ months under both p -specifications. The *level* moves: fair value drops a near-uniform $\sim \$0.30$ ($C(0) : 5.13 \rightarrow 4.77$, peak $5.47 \rightarrow 5.21$), so the call is cheap versus the **\$4.30** mid by $\sim 11\%$ (real) rather than $\sim 19\%$ (naive). The directional edge is real but smaller once the level is *earned* rather than assumed.

Caveats. The readout-arrival law $f(\tau)$ here is still the assumed gamma (only $p(\tau)$ was replaced); a fully consistent build would also draw $f(\tau)$ from the ABC 80th-death distribution. The analysis-lag component of τ is HR-uninformative, so the *effective* slope against announcement time is modestly shallower than the $+0.0275$ measured against death time (regression dilution).

Fully consistent: timing *and* outcome from one ABC pass

The section above still borrowed the assumed gamma for *when* the readout lands. The honest endpoint draws both from a single ABC pass: each accepted draw carries its own success outcome *and* its 80th-death month; the announcement is that death time plus an analysis lag ($\sim$ gamma, mean 2 mo). We then price the call directly off the empirical joint — no assumed timing law, no separately fitted $p(\tau)$ — which also captures the strong-effect tail where the 80th death slips past the Jan-2027 expiry.



Three things change, and all trace to one cause — **timing and outcome are now coupled**:

- **The hump collapses.** Peak **\$3.71** at **0.75** mo (economically flat, then decay) versus **\$5.47** at **4** mo. The 80th death is *imminent* — median **1.4** months away, since only two more deaths are needed — so rising- p has almost no window to act before the catalyst resolves.
- **Coupling cancels the lift.** A strong effect *delays* the deaths, so the high-success draws are precisely the ones that risk slipping past the Jan-2027 expiry (

$P(\text{slip}) = 14.3\%$). In the gamma model timing was independent of outcome, so rising- p lifted the price unopposed; here the lift is offset by the correlated slip-risk. **The earlier hump was partly an artifact of that independence.**

- **The edge reverses.** $C(0) = \$3.70$ versus the **\$4.30** market mid: against a fixed $U = \$20$ the call is $\sim 14\%$ *rich*, not cheap. The **11--19%** “cheapness” of the simpler models was an artifact of assumed-independent timing and the higher assumed p -level.

What this does and does not settle. It does *not* prove the call is rich: fair value scales with the payoff $U - K$. At $U = \$25$ (a blockbuster success print) the payoff is **\$15** and fair value $\approx \$5.5 > \4.30 . So the call's fairness is a bet on the *success-state price* U — exactly the HR-independent “ U view” the relative-value note isolated. And the **14.3%** slip assumes a *strictly event-driven* final analysis; if REGAL takes a date-driven cut when deaths stall, slip-risk falls and fair value climbs back toward the gamma-model range ($\sim \$4.5\text{--}4.8$).

specification	$C(0)$	peak	vs market \$4.30
naive (linear p , gamma timing)	\$5.13	\$5.47 @ 4.0 mo	$\sim 19\%$ cheap
real $p(\tau)$, gamma timing	\$4.77	\$5.21 @ 4.0 mo	$\sim 11\%$ cheap
fully consistent (timing + outcome)	\$3.70	\$3.71 @ 0.8 mo	$\sim 14\%$ rich*

*at fixed $U = \$20$ and strict event-driven analysis; reverses to cheap if $U \gtrsim \$24$ or a date-driven final cut caps the slip-risk.

The PoS is not identifiable: bounding the confound

Quoting a single readout PoS (the ≈ 0.48 above) was itself an error. Two re-sourced facts force a reckoning. **(a)** The death tail is hard data — 72 events by Dec-26-2025, 78 by May-12-2026 — and it is *incompatible* with a ~ 16 -month pooled median (a 14-month median would have produced ~ 110 deaths by now); the tail *forces* a long pooled median ($\sim 26\text{--}28$ mo). **(b)** The natural-history benchmark for untransplanted CR2 AML is ~ 8 months, yet the same tail forces the *realized* control arm to $\sim 16\text{--}22$ months in every fit — confirming the sponsor's “ $\sim 2\times$ longer than anticipated,” but for the *live, selected* population, not the textbook.

What blinded public data *cannot* do is split that long survival into baseline vs GPS effect. Sweeping the BAT anchor (the unidentifiable nuisance) bounds it:

BAT prior (assumption)	PoS	median HR	BAT-arm mOS	GPS-arm mOS
A — [8–11] natural-history (bull)	0.83	0.49	16	44
B — [8–13] NH + selection	0.77	0.52	17	42
C — [11–15] moderate selection	0.69	0.55	19	40
D — [15–20] heavy selection	0.51	0.63	21	37
E — [7–24] flat (bear)	0.43	0.67	22	36

PoS spans 0.43–0.83; median HR ≤ 0.67 in every row; the BAT-arm median never drops below 16 regardless of prior. The clinically-defensible center (modern HMA+venetoclax on a selected, transplant-deferred population, BAT ~12–18) lands at **PoS ~0.55–0.65**. The honest output is this *range*, not the retired 0.48 — and it is wider on the downside, because the generator attributes the long-survivor tail to GPS; a symmetric population tail would push HR→1.

Pricing the call off *each regime's own* accepted-draw joint (so the timing-outcome coupling stays honest — bullish, low-HR regimes read out later and slip more) gives fair value across the PoS band ($U = \$20$; market Jan-2027 \$10-call mid ~\$4.30):

regime (PoS)	$C(0)$ Jan-2027 (slip)	$C(0)$ Jan-2028 (slip)
A — bull, 0.83	\$6.48 (19%)	\$8.17 (0%)
B — 0.77	\$6.12 (17%)	\$7.57 (0%)
C — 0.69	\$5.63 (15%)	\$6.83 (0%)
D — 0.51	\$4.14 (14%)	\$5.02 (0%)
E — bear, 0.43	\$3.49 (14%)	\$4.28 (0%)

The \$4.30 market sits at regime D — **the market prices the bearish end, implied PoS ≈ 0.52 – 0.56 at $U = \$20$** . Positive EV reduces to one rule: **PoS $\times (U - K) > \sim 5.6$** — satisfied by any center-or-bull view at $U \geq \$20$. Jan-2028 removes the 14–19% expiry-slip in every regime (0%), and the recovery is largest in the bull rows, because the bull case is a late readout. The actionable asymmetry is not a mispricing but that the market has taken the bearish side of an unidentifiable confound.

Conclusion

A falling hazard ratio converts an otherwise monotonically-decaying catalyst call into one that **appreciates until** $q(t) = \alpha/(\alpha + p h)$ — *but only when the readout timing is independent of the outcome*. That assumption is what produces the ~ 4 -month no-bleed hump. Once the ABC couples timing to outcome — a stronger effect delays the deaths, so the very draws that would win are the ones that risk expiring before they read out — the lift is canceled, the hump collapses to a gentle decay, and the call's apparent cheapness ($\sim 11\text{--}19\%$) reverses to roughly fair-to-rich at $U = \$20$.

The model's verdict is therefore **conditional, not categorical**: the trade is attractive only if you believe the success-state price $U \gtrsim \$24$, or that a date-driven final analysis caps the **14%** expiry-slip risk. The robust, assumption-free takeaways are narrower: (i) the $q^* = \alpha/(\alpha + p h)$ cancellation law holds whenever timing is exogenous; (ii) the public blinded data *cannot identify* the PoS — it is bounded $\approx 0.43\text{--}0.83$ (center $\sim 0.55\text{--}0.65$), the width being the irreducible BAT $\leftrightarrow$ HR confound; and (iii) the dominant risk to a Jan-2027 call is not theta but a *late readout* — which, perversely, is most likely exactly when the drug works.